

Report

Part A1 — Data Cleaning Decisions and My Thoughts

The raw data was large and full of missing values. To make K-Means cluster clearly and keep our Streamlit app fast, I cleaned the dataset using four main steps:

- **Dropping Missing Locations:** I removed rows missing `geo_point_2d`. We need to map these businesses and run spatial clustering later, so data without coordinates is unusable.
- **Filling Missing Metrics:** For columns like `numberofemployees` and `feepaid`, I filled blanks with 0 instead of deleting the rows. This keeps our dataset large and protects our sample size.
- **Filtering for Active Businesses:** I kept only rows where the status is "Issued". "Pending" or "Cancelled" licenses are not open storefronts, so filtering helps us focus purely on active businesses.
- **Consolidating Categories:** The data had ~94 business types, which is too sparse for K-Means. I merged `business_type` and `business_subtype` for a sharper label, kept the top 15 categories, and grouped the rest as "Other" to keep the app responsive.

Part A2 — Location Clustering and Algorithm Comparison

How K-Means Maps onto Vancouver Geography

The K-Means model splits Vancouver into very clean, geometric regional blocks based on straight-line distances. It clearly isolates the high-density Downtown Core into its own group while separating the West Side and East Side into distinct quadrants. These boundaries look a lot like official municipal zones or voting districts because the algorithm partitions the space evenly.

K-Means vs. DBSCAN: Where They Agree and Disagree

Both algorithms easily agree on identifying the Downtown Core because its extreme business density stands out clearly to any model. However, they disagree completely in sparse suburban or residential neighborhoods. K-Means forces these isolated businesses

into the nearest large regional cluster, whereas DBSCAN isolates them completely and flags them as unclustered data.

Why They Disagree

This disagreement happens because K-Means assumes clusters are round and equal in size, forcing every single point into a group. On the other hand, DBSCAN looks purely at density and forms organic, irregular shapes along major commercial transit corridors like Broadway or Main Street. The "noise" points in DBSCAN represent isolated home-based businesses or quiet neighborhood shops that sit far away from busy commercial strips.

Part A3 — Feature-Based Clustering and My Thoughts

Looking at the final PCA plot, I think clustering businesses by how they operate—using size, industry, and lifecycle—shows a much more interesting story than just looking at where they are on a map.

What the Resulting Clusters Seem to Represent

I think the clusters do a great job of grouping businesses by their actual operational footprint. For example, Cluster 3 (Grey) stretches way up high, and I think this represents giant corporate offices, big retail dealers, or headquarters with huge employee numbers and high fees. Cluster 2 (Yellow) seems to capture our typical local storefronts, like neighborhood restaurants and shops that run year-round with a standard small-business size. Finally, I think Clusters 0 and 1 (Green/Blue) represent home-based businesses, short-term rentals, or seasonal services because they all share very low employee counts and similar renewal cycles.

How It Differs From Pure Geographic Clustering (A2)

I think this feature-based model is the complete opposite of our geographic clustering because it completely ignores physical distance. In Part A2, a tiny coffee shop and a massive corporate office were lumped into the same cluster just because they shared a street corner Downtown. Here, I think it is amazing that a massive department store downtown is clustered with a giant supermarket in South Vancouver because their business scales are identical. It shifts our focus from *where* a business sits to *how* it actually runs, grouping similar business behaviors across opposite sides of the city.

B4 — Cluster membership and interpretation

My Analysis of the Neighborhood Groupings

Does this clustering layout make sense?

Looking at the neighborhood assignments, I think the groupings are highly meaningful and match real-world Vancouver geography perfectly. For instance, I noticed that high-density commercial hubs like Downtown and Fairview are consistently paired together because they share massive office and retail footprints. Meanwhile, quieter residential areas like Dunbar-Southlands and Shaughnessy get grouped into their own profile due to a high density of home-based local services and construction contractors.

Were there any surprising neighborhood matches?

One grouping that surprised me was seeing Mount Pleasant and Grandview-Woodland grouped so closely with some of my standard suburban zones. I think this happened because, despite their trendy shopping districts, both neighborhoods actually host a massive number of small, home-based creative businesses and local trade contractors. This proves that the model looks deeper than just a neighborhood's main street reputation and captures the quiet, home-based economy.

How do the business distributions explain these groups?

I think the top five business distributions explain the cluster boundaries perfectly because my K-Means model was trained directly on these percentage shares, not on geographic coordinates. Here is why those top five categories make the cluster divisions so clear